# Matthew Joseph Rossi

https://www.mjrossi.com

## SUMMARY

First-year urban mobility master's student, bringing thirteen years of experience designing large-scale, high-reliability systems in regulated, multi-stakeholder environments. My expertise spans complex systems analysis, long-term planning, stakeholder coordination, and data-informed decision-making.

## EDUCATION

**EIT Urban Mobility - Sustainable Urban Mobility Transitions (SUMT)**
*M.Sc., in progress — expected 2028*
**Year 1**: IST Técnico Lisboa, Lisbon, Portugal
**Year 2**: UPC BarcelonaTech, Barcelona, Spain

**Wayne State University, Detroit, MI, USA**
*Bachelor of Science (B.S.) in Computer Science*
***Graduation:*** *May 2012*
***GPA:*** *3.41 / 4.0*
***Dean's List:*** *Winter 2009, Winter 2011, Winter 2012*
***Academic Strengths & Focus:*** *Systems analysis, software design, complex problem-solving, human–computer interaction, quantitative reasoning, data structures & algorithms*

## WORK HISTORY

**Heroku (Salesforce), New York, NY, USA** **June 2021 – May 2026**
***Senior Software Engineer***
*High-availability, high-impact technical environment*

- Designed, built, and maintained Heroku's new secure credential and authorization management system supporting multiple internal platforms and cross-functional teams within a next-generation data synchronization environment.
- Architected how interdependent microservices powered the next-generation data synchronization platform, with emphasis on maintainability, traceable ownership, and scalability.
- Developed high-level architecture and detailed design specifications for systems, presenting these findings to executive leadership, product stakeholders, and engineering teams to guide platform-wide decisions.
- Led efforts to improve team operational processes to enhance long-term sustainability, reliability, and on-call health in a high-availability environment.
- Collaborated with customers and internal teams, including engineers, architects, and product designers on support issues, system design, service integrations, and cross-team initiatives.

**Capital One, New York, NY, USA** **April 2019 – June 2021**
***Senior Software Engineer***
*Heavily regulated financial environment*

- Developed an internal underwriting and portfolio-management system under regulatory and financial constraints for the Commercial Banking division.

- Worked closely with product stakeholders on gathering product and technical requirements, and architecting test strategies to ensure software reliability for analyzing commercial underwriting risk and compliance.
- Communicated across product, business, and technical teams to evaluate trade-offs and align decisions with organizational objectives.
- Provided mentorship to junior engineers on new feature development, technical workshop presentation, and routine pair programming.

**Braintree, Chicago, IL, USA** **April 2016 – March 2019**
***Software Engineer***
*Mobile/web-based payments & financial technology sector*

- Developed new features in a test-driven development style to the Braintree payment processing platform.
- Led a cross-functional team (~20 people) to design and deliver an internal engineering conference for ~200 participants, supporting organization-wide learning and technical knowledge sharing.
- Worked as a tech lead for a group of software development apprentices on the fundamentals of object-oriented programming, and SOLID (Single Responsibility, Open/Closed, Liskov Substitution, Interface Segregation, Dependency Inversion) principles.

**Clarity Consulting, Chicago, IL, USA** **July 2013 – March 2016**
***Product Development Engineer***
*Clarity Connect, a Microsoft Skype-for-Business contact center platform*

- Led the design and build of integration points connecting the product to third-party customer relationship management (CRM), quality management, and workforce management platforms — coordinating across independently-owned systems with differing data models and release cycles.
- Delivered feature development across the product, and built and maintained the installation and deployment process.

**Ford Motor Company, Dearborn, MI, USA** **2010 – 2013**
*Open Source Coordinator (2012–2013) · Software Engineering Intern (Summers 2010, 2011)*
Automotive manufacturing; enterprise IT and legal operations

- Embedded as the engineering voice on a team of attorneys handling IT contracts and software licensing, translating technical constraints into terms a legal function could act on, and legal obligations into guidance engineering teams could follow.
- Built internal tooling on the Digital Worker team to help employees make better use of enterprise technology services (2011); Windows Server engineering (2010).

URBAN MOBILITY ENGAGEMENT

Self-directed engagement with urban mobility systems, transport policy, and civic technology:

- Founded and actively maintains the **Urbanist Atlas** (urbanistatlas.com), an open directory cataloging 146 transit and safe-streets advocacy organizations across 80 regions of the US and Canada, searchable by postal code. Built the full stack (Go API, React frontend, graph-based regional data model) and released it under open licenses: Apache 2.0 code, ODbL 1.0 data, CC BY-SA 4.0 documentation.

- Write **The Urbanist Lexicon** (mjrossi.com/blog), a publication on transport systems, infrastructure, and the transition from software engineering to urban mobility.
- Built a Ruby library integrating with a transit authority's real-time bus status API to improve data accessibility, alongside exploratory analysis of agency open data feeds.
- Participated in civic-tech and urbanism communities, including Chicago's weekly ChiHackNight.
- Member of the Rail Passengers Association, Riders Alliance (NYC), and Transportation Alternatives.

SKILLS

Systems & Analysis

- Complex systems analysis, architecture, and design
- Requirements gathering and translation
- Evaluating trade-offs in constrained, multi-stakeholder environments
- Communicating benefits and drawbacks of technical decisions
- Experience working with public open data APIs (e.g., GTFS and GTFS-realtime feeds)

Technical Skills

- Go, C#, Ruby, Python, Java, JavaScript
- Distributed systems architecture
- API design & integration
- SQL / relational databases
- Data modeling & analysis

Professional Skills

- Cross-functional communication
- Stakeholder alignment
- Mentorship and technical leadership